MATH 463 - INTEGRAL EQUATIONS

LESSON 3

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OUTLINE

1. METHODS OF SOLUTIONS TO THE FREDHOLMS INTEGRAL EQUATION
   - Methods of solutions to the Fredholm's Integral Equation with a Separable Kernel
   - Methods of solutions to the Fredholm's Integral Equation with a Degenerate Kernel
   - Eigen Values of a Fredholm's Integral Equation
   - Method of Successive Approximation to the free term
   - Method of Successive Approximation to the Kernel

METHODS OF SOLUTIONS TO THE FREDHOLM'S INTEGRAL EQUATION WITH A SEPARABLE KERNEL

Exact solution:

$$u(x) = f(x) + \lambda \int_a^b k(x,t)u(t)\ dt$$

$$k(x,t) = g(x)h(t)$$

$$u(x) = f(x) + \lambda \int_a^b g(x)h(t)u(t)\ dt$$

$$u(x) = f(x) + \lambda g(x) \int_a^b h(t)u(t)\ dt$$

$$u(x) = f(x) + \lambda g(x)c \qquad where\ c = \int_a^b h(t)u(t)\ dt$$

$$\implies u(t) = f(t) + \lambda g(t)c$$

$$\implies c = \int_a^b h(t)[f(t) + \lambda g(t)c]\ dt$$

Solve for c and put it back into u(x) to complete the solution

Example: Find the exact solution to the integral equation below

$$y(x) = -\int_0^1 xty(t)\ dt + x^2$$

Solution:

$$y(x) = x^2 - x\int_0^1 ty(t)\ dt$$

$$y(x) = x^2 - xc \qquad \textit{where } c = \int_0^1 ty(t)\ dt$$

$$\implies y(t) = t^2 - tc$$

$$\implies c = \int_0^1 t[t^2 - tc]\ dt = \frac{3}{16}$$

$$y(x) = x^2 - \frac{3x}{16}$$

METHODS OF SOLUTIONS TO THE FREDHOLM'S INTEGRAL EQUATION WITH A DEGENERATE KERNEL

Exact Solution:

$$u(x) = f(x) + \lambda \int_a^b k(x,t)u(t)\ dt$$

$$k(x,t) = \sum_{i=1}^{n} g_i(x)h_i(t)$$

$$u(x) = f(x) + \lambda \int_a^b [g_1(x)h_1(t) + g_2(x)h_2(t) + ... + g_n(x)h_n(t)]u(t)\ dt$$

$$u(x) = f(x) + \lambda g_1(x) \int_a^b h_1(t)u(t)\ dt + ... + \lambda g_n(x) \int_a^b h_n(t)u(t)\ dt$$

$$u(x) = f(x) + \lambda g_1(x)c_1 + \lambda g_2(x)c_2 + ... + \lambda g_n(x)c_n \qquad (1)$$

$$u(t) = f(t) + \lambda g_1(t)c_1 + \lambda g_2(t)c_2 + ... + \lambda g_n(t)c_n$$

Use the value of u(t) to solve for the value of all c, then put it back into (1) to complete the solution.

Example:

Solve

$$y(x) = \frac{x^2}{2} + \int_0^1 (x+t)y(t)\ dt$$

Solution:

$$y(x) = \frac{x^2}{2} + \int_0^1 xy(t)\ dt + \int_0^1 ty(t)\ dt$$

$$y(x) = \frac{x^2}{2} + x\int_0^1 y(t)\ dt + \int_0^1 ty(t)\ dt$$

$$y(x) = \frac{x^2}{2} + xc_1 + c_2$$

$$y(t) = \frac{t^2}{2} + tc_1 + c_2$$

$$c_1 = \int_0^1 y(t)\ dt$$

$$c_1 = \int_0^1 \frac{t^2}{2} + tc_1 + c_2\ dt$$

$$c_1 = \frac{1}{6} + \frac{c_1}{2} + c_2$$

$$c_1 = 2c_2 + \frac{1}{3} \qquad (1)$$

$$c_2 = \int_0^1 t[\frac{t^2}{2} + tc_1 + c_2]\, dt$$

$$c_2 = \frac{1}{8} + \frac{c_1}{3} + \frac{c_2}{2}$$

$$c_1 = \frac{3c_2}{2} - \frac{3}{8} \qquad (2)$$

Equating equation (1) and (2)

$$2c_2 + \frac{1}{3} = \frac{3c_2}{2} - \frac{3}{8}$$

$$c_2 = -\frac{17}{12}, \qquad c_1 = -\frac{23}{16}$$

Hence

$$y(x) = \frac{x^2}{2} - \frac{23x}{16} - \frac{17}{12}$$

FINDING THE EIGEN VALUES OF HOMOGENEOUS FREDHOLM'S INTEGRAL EQUATION

$$u(x) = \lambda \int_a^b k(x,t)u(t)\ dt$$

For a separable kernel, the solution to the above integral equation is $u(x) = \lambda g(x)c$ where $c = \int_a^b h(t)u(t)\ dt$

$c = 0$ gives rise to the trivial solution. $k(x,t) \neq 0$.

The eigen values are the values of λ for which $u(x) \neq 0$

Example: Find the eigen value and the corresponding eigen function of the integral equation

$$y(x) = \lambda \int_0^1 e^{x+t} y(t)\ dt$$

Solution:

$$\lambda \int_0^1 e^{x+t} y(t)\ dt = y(x)$$

$$\lambda e^x \int_0^1 e^t y(t)\ dt = y(x)$$

$$\lambda e^x c = y(x) \qquad \textit{where } c = \int_0^1 e^t y(t)\ dt$$

$$\lambda e^t c = y(t)$$

$$c = \int_0^1 e^t [\lambda e^t c]\ dt$$

$$c = \lambda c \left[\frac{e^2 - 1}{2} \right]$$

Hence the eigen value is

$$\lambda = \frac{2}{e^2 - 1}$$

The eigen function is

$$y(x) = \lambda e^x c = \frac{2e^x c}{e^2 - 1}$$

METHOD OF SUCCESSIVE APPROXIMATION TO THE FREE TERM

Approximated Solution:

$$y(x) = f(x) + \lambda \int_a^b k(x,t)y(t)\ dt \tag{1}$$

Let the solution of the above F.I.E. be given as

$$y(x) = \phi_0(x) + \lambda^1\phi_1(x) + \lambda^2\phi_2(x) + ... + \lambda^n\phi_n(x) \tag{2}$$

$$\implies y(t) = \phi_0(t) + \lambda^1\phi_1(t) + \lambda^2\phi_2(t) + ... + \lambda^n\phi_n(t) \tag{3}$$

putting 3 into 1:

$$y(x) = f(x) + \lambda \int_a^b k(x,t)[\phi_0(t) + \lambda^1\phi_1(t) + \lambda^2\phi_2(t) + ... + \lambda^n\phi_n(t)]\ dt$$

$$y(x) = f(x) + \lambda \int_a^b k(x,t)\phi_0(t) + \lambda^1 k(x,t)\phi_1(t) + \lambda^2 k(x,t)\phi_2(t) + \\ ... + \lambda^n k(x,t)\phi_n(t)\ dt$$

$$y(x) = f(x) + \lambda \int_a^b k(x,t)\phi_0(t)\ dt + \lambda^2 \int_a^b k(x,t)\phi_1(t)\ dt +$$

$$\ldots + \lambda^n \int_a^b k(x,t)\phi_{n-1}(t)\ dt + \lambda^{n+1} \int_a^b k(x,t)\phi_n(t)\ dt \qquad (4)$$

comparing 4 and 2

$$\phi_0(x) = f(x)$$

$$\phi_1(x) = \int_a^b k(x,t)\phi_0(t)\ dt$$

$$\phi_2(x) = \int_a^b k(x,t)\phi_1(t)\ dt$$

$$\phi_n(x) = \int_a^b k(x,t)\phi_{n-1}(t)\ dt$$

$$\phi_{n+1}(x) = \int_a^b k(x,t)\phi_n(t)\ dt$$

Which completes the solution.

NOTE:

1. For $y(x)$ to hold as a solution to the F.I.E, the kernel $k(x,t)$ and the free term $f(x)$ must be bounded within the limits of integration. Thus $k(x,t)$ and $f(x)$ must be continuous within the limit of integration.
2. For the series to converge, the absolute value of the common ratio should be less than 1.

Example: Using the method of successive approximation to the free term, find the solution of the F.I.E below.

$$y(x) + \int_0^1 xty(t)\ dt = x^2$$

Solution:

$$y(x) = x^2 - \int_0^1 xty(t)\ dt$$

$$f(x) = x^2, \qquad \lambda = -1, \qquad k(x,t) = xt$$

$$y(x) = \phi_0(x) + \lambda^1\phi_1(x) + \lambda^2\phi_2(x) + \ldots.. + \lambda^n\phi_n(x)$$

$$\phi_0 = f(x) = x^2$$

$$\phi_1(x) = \int_0^1 k(x,t)\phi_0(t)\ dt = \int_0^1 xt^3\ dt = \frac{x}{4}$$

$$\phi_2(x) = \int_0^1 k(x,t)\phi_1(t)\ dt = \int_0^1 \frac{xt^2}{4}\ dt = \frac{x}{12}$$

$$\phi_3(x) = \int_0^1 k(x,t)\phi_2(t)\ dt = \int_0^1 \frac{xt^2}{12}\ dt = \frac{x}{36}$$

$$\phi_4(x) = \int_0^1 k(x,t)\phi_3(t)\ dt = \int_0^1 \frac{xt^2}{36}\ dt = \frac{x}{108}$$

$$y(x) = x^2 - \frac{x}{4} + \frac{x}{12} - \frac{x}{36} + \frac{x}{108} + \ldots\ldots$$

$$common\ ratio(r) = \frac{x}{12} \times \frac{-4}{x} = \frac{-x}{36} \times \frac{12}{x} = \left|\frac{1}{3}\right|$$

Since the absolute value of the common ratio is less than 1, the series will converge.

Note: The first term is $\frac{-x}{4}$ and not x^2

$$sum\ of\ a\ G.P = \frac{a}{1-r} = \frac{-x}{4} \times \frac{1}{1-\frac{-1}{3}} = \frac{-3x}{16}$$

$$\implies y(x) = x^2 - \frac{3x}{16}$$

METHOD OF SUCCESSIVE APPROXIMATION TO THE KERNEL

Approximated Solution:

$$y(x) = f(x) + \lambda \int_a^b k(x,t)y(t)\ dt \tag{1}$$

Let the solution of the above F.I.E. be given as

$$y(x) = \phi_0(x) + \lambda^1\phi_1(x) + \lambda^2\phi_2(x) + ... + \lambda^n\phi_n(x) \tag{2}$$

$$\phi_0(x) = f(x),\ \ \phi_0(t) = f(t) \tag{3}$$

$$\phi_1(x) = \int_a^b k(x,t)\phi_0(t)\ dt \tag{4}$$

$$\phi_1(s) = \int_a^b k(s,t)\phi_0(t)\ dt$$

$$also\ \phi_2(x) = \int_a^b k(x,t)\phi_1(t)\ dt$$

$$\phi_2(s) = \int_a^b k(s,t)\phi_1(t)\ dt$$

$$\phi_2(x) = \int_a^b k(x,s)\phi_1(s)\ ds$$

$$\phi_2(x) = \int_a^b k(x,s)\left[\int_a^b k(s,t)\phi_0(t)\ dt\right]\ ds$$

$$\phi_2(x) = \int_a^b f(t)\left[\int_a^b k(x,s)k(s,t)\ ds\right]\ dt$$

$$\phi_2(x) = \int_a^b k_2(x,t)f(t)\ dt \tag{5}$$

where

$$k_2(x,t) = \int_a^b k(x,s)k_1(s,t)\ ds$$

$$\textit{also}\ \phi_3(x) = \int_a^b k(x,t)\phi_2(t)\ dt$$

$$\phi_3(s) = \int_a^b k(s,t)\phi_2(t)\ dt$$

$$\phi_3(x) = \int_a^b k(x,s)\phi_2(s)\ ds$$

$$\phi_3(x) = \int_a^b k(x,s)\left[\int_a^b k_2(s,t)f(t)\ dt\right]\ ds$$

$$\phi_3(x) = \int_a^b f(t)\left[\int_a^b k(x,s)k_2(s,t)\ ds\right]\ dt$$

$$\phi_3(x) = \int_a^b k_3(x,t)f(t)\ dt \tag{6}$$

where

$$k_3(x,t) = \int_a^b k(x,s)k_2(s,t)\ ds$$

$$\phi_4(x) = \int_a^b k_4(x,t)f(t)\ dt \tag{7}$$

where

$$k_4(x,t) = \int_a^b k(x,s)k_3(s,t)\ ds$$

$$\phi_n(x) = \int_a^b k_n(x,t)f(t)\ dt \tag{8}$$

where

$$k_n(x,t) = \int_a^b k(x,s)k_{n-1}(s,t)\ ds$$

put (3),(4),(5),(6),(7),(8) into (2)

$$y(x) = \phi_0(x) + \lambda^1\phi_1(x) + \lambda^2\phi_2(x) + \ldots.. + \lambda^n\phi_n(x)$$

$$y(x) = f(x) + \lambda\int_a^b k_1(x,t)f(t)\ dt + \lambda^2\int_a^b k_2(x,t)f(t)\ dt + \ldots..+$$

$$\lambda^n\int_a^b k_n(x,t)f(t)\ dt$$

$$y(x) = f(x) + \lambda\int_a^b [k_1(x,t) + \lambda k_2(x,t) + \ldots.. + \lambda^{n-1}k_n(x,t)]f(t)\ dt$$

$$y(x) = f(x) + \lambda\int_a^b \Gamma(x,t,\lambda)f(t)\ dt$$

$$resolvent\ \ kernel = \Gamma(x,t,\lambda) = \sum_{i=1}^{n}\lambda^{i-1}k_i(x,t)$$

Example: Using the method of successive approximation to the kernel, find the solution of the F.I.E below.

$$y(x) + \int_0^1 xty(t)\ dt = x^2$$

Solution:

$$y(x) = x^2 - \int_0^1 xty(t)\ dt$$

$$f(x) = x^2, \qquad \lambda = -1, \qquad k(x,t) = xt$$

$$y(x) = f(x) + \lambda \int_a^b \Gamma(x,t,\lambda) f(t)\ dt$$

$$k_1(x,t) = xt$$

$$k_2(x,t) = \int_0^1 k(x,s)k_1(s,t)\ ds = \int_0^1 xs^2 t\ ds = \frac{xt}{3}$$

$$k_3(x,t) = \int_0^1 k(x,s)k_2(s,t)\ ds = \int_0^1 \frac{xs^2 t}{3}\ ds = \frac{xt}{9}$$

$$k_4(x,t) = \int_0^1 k(x,s)k_3(s,t)\ ds = \int_0^1 \frac{xs^2 t}{9}\ ds = \frac{xt}{27}$$

$$\Gamma(x,t,\lambda) = k_1(x,t) - k_2(x,t) + k_3(x,t) - k_4(x,t) + ...$$

$$\Gamma(x,t,\lambda) = xt - \frac{xt}{3} + \frac{xt}{9} - \frac{xt}{27} + ...$$

$$common\ ratio(r) = \frac{-xt}{3} \times \frac{1}{xt} = \frac{xt}{9} \times \frac{-3}{xt} = \left|\frac{1}{3}\right|$$

Since the absolute value of the common ratio is less than 1, the series will converge.

$$sum\ of\ a\ G.P = \frac{a}{1-r} = \frac{xt}{1-\frac{-1}{3}} = \frac{3xt}{4}$$

$$\Gamma(x,t,\lambda) = \frac{3xt}{4}$$

$$y(x) = x^2 - \int_0^1 \frac{3xt}{4} t^2\ dt$$

$$\implies y(x) = x^2 - \frac{3x}{16}$$

Exercises

1. Find the resolvent kernel, radius of convergence of the series and the solution of the following equations by using an approximation the kernel.
   1. $$y(x) = \frac{5}{6}x + \frac{1}{2}\int_0^1 xty(t)\ dt$$
   2. $$y(x) = x + \frac{1}{2}\int_0^1 ty(t)\ dt$$
2. Find the solution of the integral equation below using an approximation to the free term
$$y(x) = \frac{11}{2}x + \frac{1}{4}\int_0^1 xty(t)\ dt$$
3. Find the eigen value (λ) of
$$y(x) = \lambda\int_0^1 xy(t) + x^9$$